A5: Angstrom Derivation and Model-Validity Plan

Assessment code: A5 (individual)
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Date: 2 November 2026 (data from the S18 live run the same day)
Repository: https://github.com/kapilnaidu1/phys39-naidu-cohen, branch main
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Markdown source: docs/assessments/a5_angstrom_derivation_plan.md

1. The rod equation and what κ and ν mean

Take a slice of rod between x and x + dx, with cross-section A = πR², perimeter P = 2πR, density ρ, specific heat c, conductivity k, and side heat-transfer coefficient h to a room at T∞. The heat stored in the slice equals the heat conducted in, minus the heat conducted out, minus the heat lost through the side:

```
ρ c A dx ∂T/∂t = kA ∂T/∂x |(x+dx) − kA ∂T/∂x |(x) − h P dx (T − T∞)
```

Dividing by ρcA dx, letting dx go to zero and writing θ = T − T∞:

```
∂θ/∂t = κ ∂²θ/∂x² − ν θ,      κ = k/(ρc),      ν = hP/(ρcA) = 2h/(ρcR)
```

κ is the thermal diffusivity, which says how fast a temperature change spreads along the rod. ν is the side-loss rate, which says how fast a slice relaxes toward room temperature through its surface. I'm assuming one temperature per cross-section (a small transverse Biot number, checked in 6.1), constant k, ρ, c and h, linear side loss, a constant room temperature, and the measured temperature at the first rod hole (T1, x = 0) as the boundary condition. At steady state the same equation becomes κθ″ = νθ, so the Module 7 decay constant is m² = ν/κ = hP/(kA).

2. Periodic solution

Drive the base as θ(0, t) = θ_b cos ωt with ω = 2π/τ, and treat the rod as semi-infinite for now (x ≥ 0, θ → 0 far away). A complex exponential is a good guess because differentiating it just gives back the same function times a constant:

```
θ~(x, t) = θ_b e^(−s x) e^(i ω t)      ∂θ~/∂t = iω θ~,      ∂²θ~/∂x² = s² θ~
```

Putting this into the rod equation and cancelling θ~ turns the PDE into algebra:

```
i ω = κ s² − ν      →      s² = (ν + i ω) / κ
```

I keep the root with a positive real part, since the other one would grow without limit, and write s = q + iq′. Then s² = q² − q′² + 2iqq′, and matching the real and imaginary parts:

```
q² − q′² = ν/κ        2 q q′ = ω/κ
(q² + q′²)² = (ν² + ω²)/κ²
q  = √[ ( ν + √(ν² + ω²) ) / (2κ) ]        q′ = √[ ( −ν + √(ν² + ω²) ) / (2κ) ]
```

The real temperature is the real part:

```
θ(x, t) = θ_b e^(−q x) cos(ω t − q′ x)
```

Every point oscillates at the drive frequency. The amplitude drops by a factor of e every 1/q, and the phase falls behind by q′x radians, which is a time delay of q′x/ω. Two limits check the algebra. With no side loss (ν = 0), q = q′ = √(ω/2κ). With side loss, q > q′, because losing heat out the side shrinks the amplitude without adding any delay. A shorter period (bigger ω) makes both q and q′ larger.

The setpoint's mean is above room temperature, so the full solution also has the steady profile from Module 7 in it. The equation is linear, so the two just add, and in the fits each sensor's constant term takes care of the mean.

For the real rod, with an insulated top a distance L above T1, the periodic solution is

```
θ~(x, t) = θ_b e^(iωt) cosh(s(L − x)) / cosh(sL) = [semi-infinite] × (1 + e^(−2s(L−x))) / (1 + e^(−2sL))
```

The extra factor is a wave reflected off the top, with a relative size of about e^(−2q(L−x)). That's what the finite-length check in 6.2 measures.

3. Inverse relations and dimensional checks

For two sensors Δx = x_2 − x_1 apart, q = ln(B_1/B_2)/Δx and q′ = Δφ/Δx. With five sensors, q and q′ are the slopes of ln B and of the unwrapped phase against x. Rearranging the two matching conditions:

```
κ = ω / (2 q q′)        ν = κ (q² − q′²)        k = ρ c κ = ρ c ω / (2 q q′)
```

So the conductivity only needs the two measured slopes, the drive frequency and ρc. You never have to measure h, which is the whole point of the method.

Quantity	Units	Check
θ, t, x	K, s, m	
$\partial\theta/\partial t$, $\kappa\partial^2\theta/\partial x^2$, $\nu\theta$	K/s each	$(m^2/s)(K/m^2) = K/s$; $(1/s)K = K/s$
s^2, q^2, q'^2	$1/m^2$	$(\nu + i\omega)/\kappa = (1/s)/(m^2/s)$
$\kappa = \omega/(2qq')$	m^2/s	$(rad/s)/(1/m^2)$
$\nu = \kappa(q^2 - q'^2)$	1/s	$(m^2/s)(1/m^2)$
$k = \rho c \kappa$	W/(m K)	$(kg/m^3)(J/(kg\ K))(m^2/s) = J/(s\ m\ K)$
$\nu = 2h/(\rho c R)$	1/s	$(W/(m^2\ K))\ /\ ((J/(m^3\ K))\ m)$

Checking against the course example (ω = 0.0628 rad/s, q = 25 m^{-1}, q′ = 18 m^{-1}): κ = 0.0628/(2 × 25 × 18) = 6.98 × 10^{-5} m^2/s and ν = 6.98 × 10^{-5} × (625 − 324) = 2.10 × 10^{-2} s^{-1}, the same as on the course page.

4. Sign and phase conventions

- x is measured from T1's hole (the first rod hole), positive going up the rod toward the top. T1 is the boundary, not the TEC thermistor or the setpoint.
- θ = T − T∞ with T∞ = 22.53 °C. That's the mean of T1 to T5 (with calibration offsets) with control off and the rod unheated, at 09:47 on 2 November.
- Each sensor is fitted as $T_i(t) = a_i + B_i \cos(\omega t + \varphi_i) + c_i(t - \bar{t})$, with $B_i > 0$. The drift term c_i soaks up the slow warming of the mean. In this convention a sensor that lags has a smaller φ_i.
- Phases are unwrapped starting at T1 and going up the rod, so $\ln B(x) = \ln B_0 - qx$ and $\varphi(x) = \varphi_0 - q'x$. Both slopes are negative, so q and q′ come out positive.
- The Arduino setpoint is $T_a + T_b \sin(\text{phase})$, which is a cosine shifted by −π/2. Only phase differences between sensors go into q′, so the shift cancels.
- The calibration offsets (from the 26 October room cross-check) are subtracted in the analysis. A constant offset only changes a_i, not B_i or φ_i.
- Errors: B_i and φ_i get their errors from the scatter when I fit each cycle separately (standard error over 5 cycles), combined with ±1 mm position uncertainty for T2 to T5 (T1 defines x = 0, so it has none). If χ^2/dof is above 1, I scale the slope errors by $\sqrt{(\chi^2/\text{dof})}$.

5. Acquisition plan

Drive: T_a = 40 °C and T_b = 16 °C (24 to 56 °C requested), PI on the TEC thermistor with K_p = 12, K_i = 0.2 and conditional-integration anti-windup, and negative commands multiplied by $r = \chi_h/\chi_c = 0.501/0.181 = 2.771$ (from Module 4) before the ±255 clamp. In our S18 comparison at 100 s, open loop got the TEC to swing 23 % of T_b, P got 76 % with a 2.6 °C droop, and PI got 87 % with no droop and no saturation, so PI is the mode for production runs. T_a and T_b stay the same for every period, so only the frequency changes.

Periods: the planning numbers below use κ and ν from the S18 finite-rod fit at 480 s (section 6.3) and the sensors at 0 to 101.6 mm.

Period (s)	1/q (mm)	predicted B5/B1	predicted lag T1 → T5 (rad)	finite-length change of B5/B1	records per cycle
100	54	0.152	1.83	+0.1 %	200
180	71	0.240	1.34	0.0 %	360
240	82	0.285	1.14	−0.8 %	480
300	90	0.319	1.01	−1.8 %	600
480	111	0.386	0.80	−3.4 %	960

The course wants the farthest sensor near B5 ≈ B1/e = 0.37 at the main period, a T1 to T5 lag of 0.5 to 2 rad, and no period under 100 s. My main period is 480 s: on S18 we measured B5/B1 = 0.385 and 0.81 rad, both on target, and the TEC followed the setpoint at 99 % amplitude. The second period is 100 s, the course minimum, where we measured B5/B1 = 0.149 and 1.79 rad. T5 still swings 0.43 °C there, about 27 times its 0.016 °C fit residual. For S20 I'm adding 240 s as a third period, in between the two, to check whether κ depends on frequency.

Sampling: one record every 0.5 s, each one the average of 1000 ADC readings (TEC) or 500 readings (each rod channel). That's 200 to 960 records per cycle, way more than a sine fit needs, and the averaging keeps

the fit residuals at 0.016 to 0.03 °C at 100 s.

Settling criterion (decided before looking at the fits): fit each complete cycle on its own. A period counts as settled from the first cycle after which every sensor's B stays within 3 % and φ within 0.05 rad of the cycle before, with no steady trend over three cycles. The mean is allowed to keep creeping, since the drift term handles that. I keep at least 5 complete settled cycles. On S18, PI at 100 s met this from its second cycle. At 480 s, the first cycle after the switch had a TEC swing 3.5 % low, so I left it out.

Run lengths for S20: about 2 cycles to settle plus 5 kept, so roughly 56 min at 480 s, 28 min at 240 s and 12 min at 100 s. That's about 1 h 40 min in one scheduled run, with the transitions between periods kept in the record and the setpoint phase continuous across them.

Sensors: TEC thermistor in the base, and T1 to T5 taped into the first five holes at 0, 25.4, 50.8, 76.2 and 101.6 mm (±1 mm, `data/module_08/sensor_map.csv`). Rod diameter 12.757 ± 0.006 mm by calipers, and L = 281.9 mm from T1's hole to the top (`data/module_08/rod_geometry.csv`). No sensor moves between periods.

Safety: the software latches off at 56 °C on T1 (the course limit for the rod base) or 65 °C on the TEC thermistor. The 70 °C thermal switch is still there as the hardware backup. Setpoints outside 10 to 56 °C are refused, and no period goes below 100 s. The highest raw readings on S18 were T1 41.3 °C and TEC 55.9 °C.

Room temperature: measure it before heating and again after the run, with everything at rest. On S18 we only measured it before. That limits how well we know the means a_i, but it doesn't affect B_i or φ_i.

6. Model-validity checks

6.1 Transverse Biot number

The 1-D equation needs each cross-section to be close to one temperature, meaning $\mathrm{Bi}_\perp = hR/(2k)$ has to be much less than 1. Since $\nu/\kappa = hP/(kA) = 2h/(kR)$, I can get it straight from the measured slopes without knowing h or k: $\mathrm{Bi}_\perp = (\nu/\kappa)R^2/4$. With R = 6.378 mm:

Source	ν/κ (m⁻²)	$\mathrm{Bi}_\perp$
Angstrom, 480 s finite-rod fit	21.7 ± 3.8	2.2×10^{-4}
Angstrom, 100 s	41.6 (ν poorly determined)	4.2×10^{-4}
Module 7 steady profile, m²	16.8	1.7×10^{-4}

All of these are under 10^{-3}, well below my tolerance of 0.01, so the temperature variation across the rod is negligible. The 480 s value of ν/κ also agrees with the separate Module 7 steady profile to within 1.3 standard errors. Using the handbook ρc for aluminum (2.42×10^6 J/(m³ K)), $\nu = 2.04 \times 10^{-3}$ s⁻¹ works out to $h \approx \nu\rho cR/2 \approx 16$ W/(m² K), which is a normal value for still air.

6.2 Finite length

Using the measured $s = q + iq'$ at each period, the finite-rod factor from section 2 changes the ratios B_i/B_1 and the lags $\varphi_1 - \varphi_i$ by:

Period		T2	T3	T4	T5
100 s	B_i/B_1	0.0 %	0.0 %	+0.03 %	+0.13 %
100 s	phase (rad)	0.000	0.000	0.000	0.000
480 s	B_i/B_1	−0.4 %	−1.0 %	−2.0 %	−3.1 %
480 s	phase (rad)	0.002	0.001	−0.003	−0.016

With a 1 % tolerance, the semi-infinite form is fine for all five sensors at 100 s, but at 480 s only T1 to T3 pass. So at 480 s I fit the insulated-tip finite-rod solution to all five sensors instead of throwing out T4 and T5.

6.3 Data-based checks (S18)

- Boundary waveform: at T1 the second harmonic is 2.3 % (100 s) and 1.5 % (480 s) of the fundamental, the heating and cooling halves agree within 0.2 °C (100 s comparison), and the PI command never hit the clamp. T1 is a clean sine.
- Sign of ν: $q > q'$ at both periods, so $\nu > 0$, which side loss requires.
- Spatial residuals: the straight-line fits have χ^2/dof of 3.3 and 4.7 (100 s) and 2.3 and 3.2 (480 s). The points scatter about twice as much as their errors, so there is a sensor-to-sensor effect of a few percent (contact or position) that isn't in the error budget yet.
- Frequency consistency, from `data/module_08/angstrom_finite_angstrom_live_20261102_094506.txt`:

Fit	κ at 100 s (10^{-5} m²/s)	κ at 480 s (10^{-5} m²/s)	difference	χ^2 at 480 s (6 dof)
semi-infinite	9.74 ± 0.23	8.93 ± 0.25	2.4 σ	16.8
finite rod, L = 281.9 mm	9.74 ± 0.23	9.40 ± 0.24	1.0 σ	11.1

The semi-infinite fits don't agree between the two periods. The finite-rod fits agree within one standard error and fit the 480 s data better. So the apparent frequency dependence is the finite-length effect that 6.2 predicts.

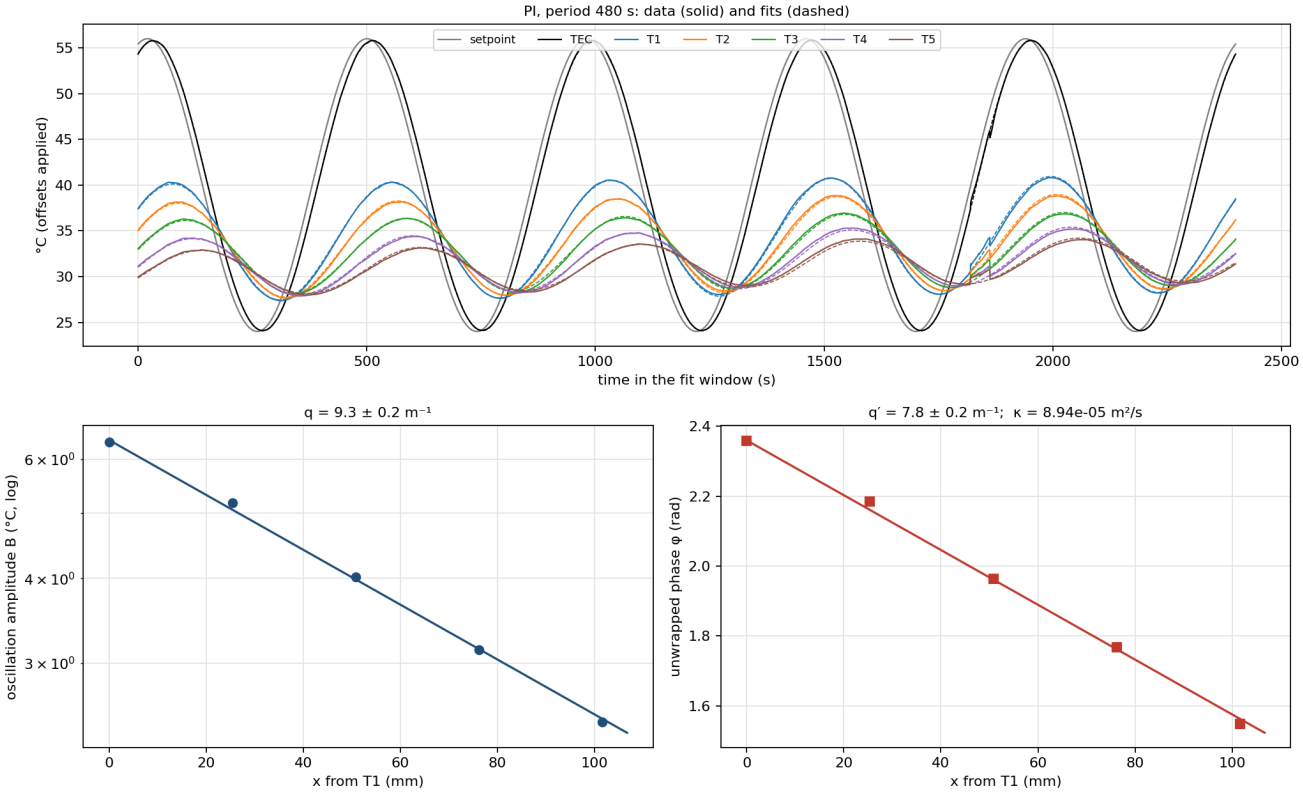


Figure 1. PI at 480 s, cycles 2 to 6 of the S18 run. Top: setpoint, TEC and T1 to T5 with each sensor's fit (dashed). Bottom: amplitude on a log scale and unwrapped phase against position, with the semi-infinite lines (q = 9.34 ± 0.18 m⁻¹, q′ = 7.84 ± 0.15 m⁻¹).

- Comparison with the literature: the handbook values for 6061-T6 (k = 167 W/(m K), ρ = 2700 kg/m³, c = 896 J/(kg K)) give κ = 6.9 × 10^{-5} m²/s, about 30 % below our finite-rod values. Neither the Biot number nor the finite length explains that. Things I want to test in S19 to S22: the sensor positions (κ goes as Δx^2, so a 1 mm error in each 25.4 mm spacing moves κ by 8 %, and the bead might not sit on the hole's center line), different response lags between the taped sensors, and the alloy itself.

7. Conclusion: do the retained sensors support the 1-D analysis?

Yes, with one condition. Across the rod, $Bi_\perp$ is at most 4 × 10^{-4}, so each cross-section is at one temperature and T1 to T5 are measuring the 1-D field. Along the rod, the semi-infinite solution works for all five sensors at 100 s (finite-length change at most 0.13 %), but at 480 s it's off by 2 to 3 % at T4 and T5. There I use the finite-rod solution with L = 281.9 mm, which brings κ at the two periods into agreement (9.74 ± 0.23 and 9.40 ± 0.24 × 10^{-5} m²/s). The remaining gap to the handbook κ, and the spatial χ^2 of 2 to 5, are open questions for the production run, not signs that the one-dimensional picture is wrong.

8. Files and commands

- Live run: `data/module_08/angstrom_live_20261102_094506.csv` (with `_events.txt`, `_notes.csv`, `_meta.json`); sketch `arduino/module_08/angstrom_tec/angstrom_tec.ino`; GUI `python/angstrom_gui.py`.
- Notes: `docs/module_notes/module_08_angstrom.md`.

```
RUN=data/module_08/angstrom_live_20261102_094506.csv
CAL=data/module_08/calibration_rod_run_20261026_112627.csv
```

```
python3 python/analysis/angstrom_fit.py $RUN --mode PI --period 100 --cycles 5 --offsets $CAL
python3 python/analysis/angstrom_fit.py $RUN --mode PI --period 480 --cycles 5 \
    --t-end 5507.21 --offsets $CAL
python3 python/analysis/angstrom_fit.py $RUN --compare --period 100 --cycles 3 --offsets $CAL
python3 python/analysis/angstrom_finite_fit.py \
    data/module_08/angstrom_fit_angstrom_live_20261102_094506_PI_100s.csv \
    data/module_08/angstrom_fit_angstrom_live_20261102_094506_PI_480s.csv \
    --length-mm 281.9 --dx-mm 1
```